The Butcher-Tree Counting Transseries

All-Order Pólya-Tree Asymptotics, Lambert–W Reversion, and the Asymptotic Inverse of A000081

A self-contained derivation prepared for Vladimir Reshetnikov

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Abstract

Let a_n be OEIS A000081, the number of unlabelled, non-plane rooted trees with n vertices. These are the rooted-tree shapes that index classical Butcher series and B-series. Their ordinary generating function

$$T(z) = \sum_{n\geq 1} a_n z^n$$

satisfies Pólya's implicit multiset equation

$$T(z) = z \exp\left(\sum_{k\geq 1} \frac{T(z^k)}{k}\right).$$

This article develops the complete coefficient calculus associated with that equation. We factor the generating function through the Cayley tree function,

$$T(z) = -W_0\left(-z \exp\left(\sum_{k\geq 2} \frac{T(z^k)}{k}\right)\right),$$

and exploit the universal square-root singularity of W_0 at $-1/e$.

The first principal result is an all-orders Puiseux expansion at the dominant singularity ρ. Every coefficient is given explicitly by a finite formula in universal Lambert coefficients, complete and partial Bell polynomials, and the derivatives of the analytic Pólya disturbance. Exact singularity transfer then gives

$$a_n \sim \rho^{-n} n^{-3/2} \sum_{k\geq 0} \frac{A_k}{n^k},$$

where every A_k has a closed finite formula involving Bernoulli polynomials and Bell polynomials. Numerically,

$$\begin{aligned} \rho &= 0.33832185689920769519611262571701705318\ldots, \\ \rho^{-1} &= 2.9557652856519949747148175241231 9\ldots. \end{aligned}$$

and

$$\begin{aligned} a_n \sim\ & 0.43992401257102530404090339143454 5\ldots\ \rho^{-n} n^{-3/2} \\ & \times \left(1 + \frac{0.10040348009640143476\ldots}{n} + \cdots\right). \end{aligned}$$

The second principal result is the full asymptotic inverse. After normalizing x by $X = (-\log\rho)x$, the inverse problem reduces to

$$Z = X - \frac{3}{2}\log X + K(X^{-1}),$$

where the coefficients of K are explicit Bell-polynomial transforms of the forward coefficients. The entire logarithmic skeleton is summed exactly by the real branch W_{-1}, and the remaining correction is an ordinary inverse power series with a Lagrange–Bürmann coefficient formula. An equivalent polynomial–logarithmic expansion is supplied with a closed recurrence for all coefficient polynomials.

Finally, we distinguish the smooth asymptotic inverse from the integer staircase inverse, derive the unavoidable periodic rounding layer, and give a contour-resolved transseries over the recursively generated singularity atlas. The first inherited singularity at $-\sqrt{\rho}$ is worked out explicitly. It produces an alternating forward sector of size $\rho^{-n/2}n^{-3/2}$ and, after inversion, a log-periodic sector of magnitude $y^{-1/2}(\log y)^{-3/4}$. Thus both the perturbative logarithmic expansion and the beyond-all-orders exponential sectors are described by general coefficient formulae.

How to read the word "full"

There are three nested senses in which an asymptotic description can be full.

1. The *dominant Poincaré sector* contains every inverse power of n multiplying $\rho^{-n}n^{-3/2}$. In this article its coefficients are explicit and unconditional.
2. The *smooth inverse sector* contains every polynomial in $\log\log y$ divided by a power of $\log y$, or equivalently every inverse power correction to an exact W_{-1} core. Its coefficients are also explicit and unconditional as formal asymptotic coefficients.
3. A *global transseries* additionally resolves exponentially smaller contributions from other singularities of analytic continuation. Since the Pólya-tree generating function has many sheets and the unit circle is a natural boundary, such sectors must be indexed by a chosen continuation sheet and contour. We give the complete recursive local calculus and the corresponding contour-resolved coefficient formula; Stokes/intersection multipliers are retained whenever they depend on that global choice.

This distinction prevents the common but serious error of calling a single finite numerical expansion a transseries, and also prevents a sheet-dependent analytic-continuation statement from being presented as globally canonical.

Contents

Part I

Combinatorics and Coefficient Calculus

Chapter 1

Butcher trees and the sequence A000081

1.1 The objects being counted

A *rooted tree* is a finite connected acyclic graph with one distinguished vertex. It is *non-plane* when the children of a vertex carry no left-to-right order, and it is *unlabelled* when two trees differing only by a relabelling of vertices are identified. Let a_n be the number of isomorphism classes of such trees on n vertices. We use

$$a_0 = 0, \qquad (a_n)_{n\geq 1} = 1, 1, 2, 4, 9, 20, 48, 115, 286, 719, 1842, \ldots.$$

This is OEIS A000081 [15, 17]. The same tree shapes occur as the combinatorial indices of Butcher series [3, 4, 5, 14]: removing the root decomposes a tree into an unordered multiset of rooted trees, while grafting the roots of a multiset of trees to a new vertex reconstructs the original tree.

The adjective "Butcher" describes the role of these shapes in numerical analysis; it does not change the enumeration. A B-series assigns a coefficient and an elementary differential to each non-plane rooted tree. Consequently a_n is the number of possible tree shapes of order n before method-dependent weights, symmetry factors, and elementary differentials are attached.

1.2 The multiset specification

Let $\mathcal{T}$ denote the combinatorial class of unlabelled non-plane rooted trees and let $\mathcal{Z}$ denote one atom. The root decomposition is

$$\mathcal{T} = \mathcal{Z} \times \mathrm{MSET}(\mathcal{T}). \tag{1.1}$$

For unlabelled objects, the multiset construction is encoded by the cycle index of the symmetric groups [18, 2]. Therefore the ordinary generating function

$$T(z) = \sum_{n\geq 1} a_n z^n \tag{1.2}$$

satisfies

$$\boxed{T(z) = z \exp\left(\sum_{k\geq 1} \frac{T(z^k)}{k}\right).} \tag{1.3}$$

The identity holds formally in $\mathbb{Z}[[z]]$; no analytic convergence is needed for its coefficientwise interpretation.

Direct derivation. A multiset containing m_j copies of each tree shape j contributes the monomial $z^{\sum_j m_j |j|}$. For one available object of size r, arbitrary multiplicity contributes $(1 - z^r)^{-1}$. Since there are a_r possible shapes of size r, the forest attached to the root has generating function

$$\prod_{r\geq 1}(1-z^r)^{-a_r}.$$

Multiplication by the root gives

$$T(z) = z\prod_{r\geq 1}(1-z^r)^{-a_r}. \tag{1.4}$$

Taking a formal logarithm and using $-\log(1-z^r) = \sum_{k\geq 1} z^{rk}/k$ yields

$$\log\frac{T(z)}{z} = \sum_{r\geq 1} a_r \sum_{k\geq 1}\frac{z^{rk}}{k} = \sum_{k\geq 1}\frac{1}{k}\sum_{r\geq 1} a_r (z^k)^r = \sum_{k\geq 1}\frac{T(z^k)}{k},$$

which is Equation (1.3). □

1.3 An exact quadratic-time recurrence

Define the divisor transform

$$b_m = \sum_{d|m} d\, a_d. \tag{1.5}$$

Then the sequence is generated exactly by the following recurrence.

Theorem 1.1 (Otter–Pólya recurrence). *For $n \geq 2$,*

$$\boxed{(n-1)a_n = \sum_{j=1}^{n-1} b_j a_{n-j}, \qquad b_j = \sum_{d|j} d\, a_d,} \tag{1.6}$$

with $a_1 = 1$.

Proof. Set $F(z) = T(z)/z = \sum_{n\geq 0} a_{n+1} z^n$. From Equation (1.4),

$$\log F(z) = \sum_{m\geq 1}\frac{b_m}{m} z^m.$$

Logarithmic differentiation gives

$$F'(z) = F(z)\sum_{m\geq 1} b_m z^{m-1}.$$

The coefficient of z^{n-1} is

$$n a_{n+1} = \sum_{j=1}^{n} b_j a_{n-j+1}.$$

Replacing n by $n-1$ proves Equation (1.6). □

Remark 1.2 (Why the recurrence is genuinely triangular). Although b_n contains a_n, the formula for a_n uses only b_j with $j < n$. After computing a_n, one may add the contribution $n a_n$ to every b_{kn}. This sieve implementation takes $O(N^2)$ integer arithmetic operations for $a_1, \ldots, a_N$, and it avoids expansion of large products.

Chapter 2

Bell polynomials, powers, and reversion

2.1 Exponential Bell polynomials

For integers $n \geq k \geq 0$, the exponential partial Bell polynomial is

$$\mathcal{B}_{n,k}(x_1, x_2, \ldots) = n! \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_j m_j = k,\ \sum_j j m_j = n}} \prod_{j \geq 1} \frac{1}{m_j!} \left(\frac{x_j}{j!} \right)^{m_j}. \tag{2.1}$$

The complete polynomial is $\mathcal{B}_n = \sum_{k=0}^{n} \mathcal{B}_{n,k}$. We shall repeatedly use the generating identities

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, x_2, \ldots) \frac{t^n}{n!}, \tag{2.2}$$

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \ldots, x_n) \frac{t^n}{n!}. \tag{2.3}$$

These are exactly the coefficient packages required by Faà di Bruno composition, exponentiation, and logarithmic conversion [6, 19].

2.2 A unit-series power formula

Let

$$U(X) = 1 + \sum_{j \geq 1} u_j X^j.$$

For arbitrary $\beta \in \mathbb{C}$, formal binomial expansion and Equation (2.2) give

$$\boxed{[X^N]\, U(X)^\beta = \frac{1}{N!} \sum_{k=0}^{N} \beta^{\underline{k}}\, \mathcal{B}_{N,k}(1!u_1, 2!u_2, \ldots),} \tag{2.4}$$

where $\beta^{\underline{k}} = \beta(\beta - 1) \cdots (\beta - k + 1)$. Every sum in this article involving a fractional power of a unit series can therefore be reduced to a finite Bell-polynomial expression.

2.3 Lagrange–Bürmann inversion

We record the form of Lagrange inversion used below [6, 12, 19].

Theorem 2.1 (Lagrange–Bürmann). *Let $q = w\phi(w)$ with $\phi(0) \neq 0$, and let $w = w(q)$ be the inverse germ with $w(0) = 0$. Then, for $m \geq 1$,*

$$[q^m]\, w(q) = \frac{1}{m}[u^{m-1}]\, \phi(u)^{-m}. \tag{2.5}$$

More generally, for a formal equation $w = t\Phi(w)$,

$$w(t) = \sum_{m\geq 1} \frac{t^m}{m!} \left.\frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}} \Phi(u)^m \right|_{u=0}. \tag{2.6}$$

Proof. Formal residue substitution gives

$$[q^m]\, w(q) = \operatorname*{Res}_{q=0} \frac{w(q)}{q^{m+1}}\, \mathrm{d}q.$$

Substitute $q = u\phi(u)$ and integrate the derivative term formally. The result is $m^{-1}[u^{m-1}]\, \phi(u)^{-m}$. Applying the same identity to $w = t\Phi(w)$ and Taylor-expanding the coefficient gives Equation (2.6). □

2.4 Logarithms as Bell transforms

If $F(t) = 1 + \sum_{j\geq 1} f_j t^j$, then

$$[t^n]\, \log F(t) = \frac{1}{n!} \sum_{k=1}^{n} (-1)^{k-1}(k-1)!\mathcal{B}_{n,k}(1!f_1, 2!f_2, \ldots). \tag{2.7}$$

This identity will convert the forward asymptotic coefficients into the additive phase coefficients needed for inversion.

Master formula

The entire calculation uses only four coefficient operations:

1. exact divisor convolution for the coefficients a_n;
2. complete Bell polynomials for exponentials and derivative jets;
3. partial Bell polynomials for arbitrary powers and logarithms;
4. Lagrange–Bürmann inversion for the Lambert branch and for the inverse counting law.

No coefficient is defined by an unnamed remainder or an implicit numerical fit.

Part II

The Dominant Forward Transseries

Chapter 3

Cayley–Lambert factorization and the dominant singularity

3.1 Separating the self-interaction

Write

$$R(z) = \sum_{k\geq 2} \frac{T(z^k)}{k}, \qquad \zeta(z) = z\mathrm{e}^{R(z)}. \tag{3.1}$$

Then Equation (1.3) becomes

$$T(z) = \zeta(z)\mathrm{e}^{T(z)}. \tag{3.2}$$

Let $C(u)$ be the Cayley tree function, characterized near $u = 0$ by $C = u\mathrm{e}^{C}$. Since $C(u) = -W_0(-u)$ [7, 13],

$$\boxed{T(z) = C(\zeta(z)) = -W_0(-\zeta(z)).} \tag{3.3}$$

The nontrivial Pólya symmetries are now confined to the analytic disturbance R; the singular mechanism itself is the universal Lambert branch point.

3.2 A useful coefficient representation of the disturbance

From the exact coefficients of T,

$$R(z) = \sum_{k\geq 2}\sum_{m\geq 1} \frac{a_m}{k} z^{km} = \sum_{r\geq 2} h_r z^r, \tag{3.4}$$

$$h_r = \frac{1}{r} \sum_{\substack{d|r \\ d<r}} d\, a_d = \frac{b_r - r a_r}{r}. \tag{3.5}$$

Unlike a_r, the coefficient h_r depends only on tree numbers with indices at most $r/2$. This is analytically decisive: R has radius of convergence $\sqrt{\rho}$, whereas T has radius ρ.

3.3 Characterization of the Otter singularity

Let ρ be the radius of convergence of T. Positivity and aperiodicity of (a_n) imply that the positive real point ρ is the unique singularity of T on $|z| = \rho$. Since R is analytic for $|z| < \sqrt{\rho}$, the first singularity in Equation (3.3) occurs when its argument reaches the Cayley critical value $1/e$. Hence

$$T(\rho) = 1, \qquad \zeta(\rho) = \frac{1}{e}, \qquad \log\rho + 1 + R(\rho) = 0. \tag{3.6}$$

The last equation is especially convenient numerically because the coefficients h_r decay geometrically after multiplication by ρ^r.

Theorem 3.1 (Dominant square-root singularity)**.** *There is a unique $\rho \in (0,1)$ satisfying Equation (3.6). In a dented neighbourhood of ρ, T has a convergent Puiseux expansion in powers of $(1-z/\rho)^{1/2}$. Its leading term is*

$$T(z) = 1 - \sqrt{2\ell_1}\left(1 - \frac{z}{\rho}\right)^{1/2} + O\left(1 - \frac{z}{\rho}\right), \qquad \ell_1 = 1 + \rho R'(\rho) > 0. \tag{3.7}$$

Proof. The classical Pólya–Otter argument gives a finite positive radius and the critical equations above [18, 17, 12]. Differentiating $\zeta = z\mathrm{e}^R$ gives $e\rho\zeta'(\rho) = 1 + \rho R'(\rho) = \ell_1 > 0$. The local inverse of $u \mapsto u\mathrm{e}^{-u}$ at its nondegenerate maximum $u = 1$ therefore has a square-root branch, equivalently W_0 has its standard square-root branch at $-1/e$. Substitution of the nonzero linear term of $1 - e\zeta(z)$ gives Equation (3.7). Aperiodicity excludes another singularity of equal modulus. □

3.4 High-precision constants

Using Equations (1.6), (3.5) and (3.6) gives

$$\rho = 0.3383218568992076951961126257170170531837746075329677955723038\ldots, \tag{3.8}$$

$$\alpha := \rho^{-1} = 2.9557652856519949747148175241231945883754923046635965953504725\ldots, \tag{3.9}$$

$$\lambda := \log\alpha = 1.0837575972446246604827119808961742790496621242026722349144271\ldots, \tag{3.10}$$

$$R(\rho) = 0.0837575972446246604827119808961742790496621242026722349144271\ldots. \tag{3.11}$$

The equality $\lambda = 1 + R(\rho)$ is an immediate rearrangement of Equation (3.6).

Chapter 4

The universal Lambert branch coefficients

4.1 A canonical square-root coordinate

Set

$$w = 1 - T, \qquad q = \sqrt{2(1 - e\zeta)}. \tag{4.1}$$

Because $e\zeta = T\mathrm{e}^{1-T} = (1-w)\mathrm{e}^{w}$,

$$\begin{aligned} q^2 &= 2\big(1 - (1-w)\mathrm{e}^{w}\big) \\ &= w^2 h(w), \qquad h(w) = \sum_{j\geq 0} \frac{2}{(j+2)j!} w^j = 1 + \frac{2}{3}w + \frac{1}{4}w^2 + \frac{1}{15}w^3 + \cdots . \end{aligned} \tag{4.2}$$

We choose the branch for which $q/w \to 1$. Then $q = w\sqrt{h(w)}$ is an ordinary invertible power series.

Theorem 4.1 (Universal Lambert coefficients)**.** *The inverse series has the form*

$$w(q) = \sum_{m\geq 1} \omega_m q^m, \qquad \boxed{\omega_m = \frac{1}{m}[u^{m-1}]\, h(u)^{-m/2}.} \tag{4.3}$$

Consequently

$$-W_0(-\zeta) = 1 - \sum_{m\geq 1} \omega_m \big(2(1 - e\zeta)\big)^{m/2}. \tag{4.4}$$

Proof. Apply Theorem 2.1 to $q = w\phi(w)$ with $\phi(w) = \sqrt{h(w)}$. This gives $\omega_m = m^{-1}[u^{m-1}]\phi(u)^{-m}$, which is Equation (4.3). Since $T = 1 - w$, substitution proves Equation (4.4). $\square$

The first universal coefficients are

$$\begin{aligned} w(q) = q &- \frac{1}{3}q^2 + \frac{11}{72}q^3 - \frac{43}{540}q^4 + \frac{769}{17280}q^5 - \frac{221}{8505}q^6 \\ &+ \frac{680863}{43545600}q^7 - \frac{1963}{204120}q^8 + \frac{226287557}{37623398400}q^9 - \frac{5776369}{1515591000}q^{10} + \cdots . \end{aligned} \tag{4.5}$$

They are rational and independent of the tree class. Every class satisfying $T = \zeta\mathrm{e}^{T}$ with analytic ζ uses exactly the same sequence (ω_m); only the expansion of $1 - e\zeta(z)$ changes.

Chapter 5

Complete Puiseux expansion at the dominant singularity

5.1 Scaled logarithmic jets

Put

$$X = 1 - \frac{z}{\rho}, \qquad L(z) = \log \zeta(z) = \log z + R(z), \tag{5.1}$$

and define the scaled jets

$$\ell_j = \rho^j L^{(j)}(\rho) = (-1)^{j-1}(j-1)! + \rho^j R^{(j)}(\rho). \tag{5.2}$$

The derivatives of R are themselves explicit convergent sums:

$$\rho^j R^{(j)}(\rho) = \sum_{m \geq j} h_m m^{\underline{j}} \rho^m. \tag{5.3}$$

Define

$$S(X) = 2\big(1 - e\zeta(\rho(1-X))\big) = \sum_{j \geq 1} s_j X^j. \tag{5.4}$$

Since $e\zeta(\rho) = 1$, complete Bell polynomials exponentiate the Taylor series of $L + 1$.

Lemma 5.1 (Disturbance coefficients). *For $j \geq 1$,*

$$\boxed{s_j = \frac{2(-1)^{j+1}}{j!} \mathcal{B}_j(\ell_1, \ell_2, \ldots, \ell_j).} \tag{5.5}$$

In particular $s_1 = 2\ell_1 = 2(1 + \rho R'(\rho)) > 0$.

Proof. Taylor expansion at $z = \rho$ gives

$$1 + L(\rho(1-X)) = \sum_{j \geq 1} (-1)^j \ell_j \frac{X^j}{j!}.$$

Use Equation (2.3). Every monomial of $\mathcal{B}_j$ has weighted degree j, so replacing ℓ_r by $(-1)^r \ell_r$ multiplies the polynomial by $(-1)^j$. Since $S = 2(1 - \exp(1 + L))$, the stated sign follows. $\square$

5.2 The all-order coefficient formula

Normalize

$$U(X) = \frac{S(X)}{s_1 X} = 1 + \sum_{j\geq 1} u_j X^j, \qquad u_j = \frac{s_{j+1}}{s_1}. \tag{5.6}$$

Then $q = S(X)^{1/2} = s_1^{1/2} X^{1/2} U(X)^{1/2}$. Write

$$T(z) = 1 + \sum_{r\geq 1} t_r X^{r/2}. \tag{5.7}$$

All Puiseux coefficients

For every $r \geq 1$,

$$\boxed{t_r = - \sum_{\substack{1\leq m\leq r \\ m\equiv r \pmod 2}} \omega_m s_1^{m/2} [X^{(r-m)/2}]\, U(X)^{m/2}.} \tag{5.8}$$

The remaining power coefficient is the finite Bell sum

$$[X^N]\, U(X)^{m/2} = \frac{1}{N!} \sum_{k=0}^{N} m/2^{\underline{k}} \mathcal{B}_{N,k}(1!u_1, 2!u_2, \ldots). \tag{5.9}$$

Thus t_r is a finite expression in $\omega_1, \ldots, \omega_r$ and $\ell_1, \ldots, \ell_{(r+1)/2}$.

Proof. Substitute $q = s_1^{1/2} X^{1/2} U^{1/2}$ into $T = 1 - \sum_m \omega_m q^m$. A term with index m can contribute to $X^{r/2}$ only if $r - m$ is a nonnegative even integer. Extracting that coefficient gives Equation (5.8), and Equation (2.4) gives Equation (5.9). □

The first few coefficients, useful for hand checks, are

$$t_1 = -\sqrt{s_1}, \tag{5.10}$$

$$t_2 = \frac{s_1}{3}, \tag{5.11}$$

$$t_3 = -\frac{11 s_1^2 + 36 s_2}{72\sqrt{s_1}}, \tag{5.12}$$

$$t_4 = \frac{43 s_1^2 + 180 s_2}{540}, \tag{5.13}$$

$$t_5 = -\frac{769 s_1^4 + 3960 s_1^2 s_2 + 8640 s_1 s_3 - 2160 s_2^2}{17280 s_1^{3/2}}, \tag{5.14}$$

$$t_6 = \frac{442 s_1^3 + 2709 s_1 s_2 + 5670 s_3}{17010}. \tag{5.15}$$

5.3 Numerical Puiseux coefficients

The first coefficients generated from Equation (5.8) are shown in Table 5.1. The displayed digits agree with the independent calculations of Genitrini and continue directly to arbitrary order.

Table 5.1: Coefficients in $T(z) = 1 + \sum_{r \geq 1} t_r (1 - z/\rho)^{r/2}$.

r	t_r
1	$-1.559490020374640885542207396$
2	$0.8106697078826992814381417462$
3	$-0.2854870216128456053038513708$
4	$0.1653723657120838934765135390$
5	$-0.3424599704021542010422002117$
6	$0.3174072259465285635680708874$
7	$-0.1077788002916310091822310895$
8	$0.06138495705583510468873149925$
9	$-0.1952123835975564636127698720$
10	$0.2059848312779074187361657348$
11	$-0.05272470849819056231533821627$
12	$0.01702656875495366944604539502$
13	$-0.1523706243663253962860946421$
14	$0.1737028832998504628939446698$
15	$-0.01447370373952704485673111250$
16	$-0.02189951761121556223267283747$
17	$-0.1445471935709097054769461137$
18	$0.1760771088850177790623322866$

Remark 5.2 (Analytic versus singular terms). The even indices $t_{2j} X^j$ are analytic at $z = \rho$. Any fixed analytic monomial contributes no Taylor coefficients once $n > j$; the algebraic coefficient asymptotics from the dominant point therefore depend only on the odd coefficients t_{2j+1}. The analytic part is not discarded globally: its continuation participates in the exponentially smaller sectors studied in Part IV.

Chapter 6

From Puiseux coefficients to the full asymptotic sequence

6.1 Exact coefficient extraction for a singular monomial

For $\beta \notin \mathbb{N}_0$,

$$[z^n]\left(1-\frac{z}{\rho}\right)^{\beta} = \rho^{-n}\frac{\Gamma(n-\beta)}{\Gamma(-\beta)\Gamma(n+1)}. \tag{6.1}$$

The ratio of gamma functions has a complete inverse-power expansion. Let $B_m(x)$ denote the Bernoulli polynomials, with $te^{xt}/(e^t-1) = \sum_{m\geq 0} B_m(x)t^m/m!$. Define

$$d_m(\beta) = \frac{(-1)^{m+1}}{m(m+1)}\big(B_{m+1}(-\beta) - B_{m+1}(1)\big), \qquad m \geq 1, \tag{6.2}$$

and let $g_r(\beta)$ be determined by

$$\exp\left(\sum_{m\geq 1} d_m(\beta)t^m\right) = \sum_{r\geq 0} g_r(\beta)t^r. \tag{6.3}$$

Equivalently,

$$g_0(\beta) = 1, \tag{6.4}$$

$$g_r(\beta) = \frac{1}{r}\sum_{m=1}^{r} m\, d_m(\beta)g_{r-m}(\beta), \tag{6.5}$$

$$g_r(\beta) = \frac{1}{r!}\mathcal{B}_r(1!d_1(\beta), 2!d_2(\beta), \ldots, r!d_r(\beta)). \tag{6.6}$$

Theorem 6.1 (Gamma-ratio transfer coefficients)**.** *For fixed β and $n \to \infty$ in a sector excluding the negative real axis,*

$$\frac{\Gamma(n-\beta)}{\Gamma(n+1)} \sim n^{-\beta-1}\sum_{r\geq 0}\frac{g_r(\beta)}{n^r}. \tag{6.7}$$

The coefficients are given by any of Equations (6.3), (6.5) and (6.6).

Proof. The Bernoulli-polynomial form of Stirling's expansion gives

$$\log\frac{\Gamma(n-\beta)}{\Gamma(n+1)} = -(\beta+1)\log n + \sum_{m\geq 1}\frac{d_m(\beta)}{n^m}.$$

Exponentiating and using the complete Bell-polynomial identity proves the claim. The recurrence follows by differentiating Equation (6.3) and comparing coefficients. □

For example,

$$\big(g_r(1/2)\big)_{r\geq 0} = 1, \frac{3}{8}, \frac{25}{128}, \frac{105}{1024}, \frac{1659}{32768}, \frac{6237}{262144}, \ldots. \tag{6.8}$$

6.2 All forward coefficients

Combining the odd Puiseux terms with Equations (6.1) and (6.7) yields the desired complete Poincaré sector.

Theorem 6.2 (Full dominant expansion of A000081). *For every fixed $K \geq 0$,*

$$a_n = \rho^{-n} n^{-3/2} \left(\sum_{k=0}^{K} \frac{A_k}{n^k} + O(n^{-K-1}) \right), \qquad n \to \infty, \tag{6.9}$$

where

$$\boxed{A_k = \sum_{j=0}^{k} \frac{t_{2j+1}}{\Gamma(-j-1/2)} g_{k-j}\left(j + \frac{1}{2}\right).} \tag{6.10}$$

Equivalently, writing $\Theta_k = \sqrt{\pi} A_k$,

$$\boxed{\Theta_k = \sum_{j=0}^{k} t_{2j+1} g_{k-j}\left(j + \frac{1}{2}\right) \frac{(2j+2)!}{(-4)^{j+1}(j+1)!}.} \tag{6.11}$$

Every coefficient is therefore a finite expression in rational universal constants, Bernoulli polynomials, Bell polynomials, and the disturbance jets $\ell_1, \ldots, \ell_{k+1}$.

Proof. The term $t_{2j+1} X^{j+1/2}$ contributes

$$\rho^{-n} \frac{t_{2j+1}}{\Gamma(-j-1/2)} n^{-j-3/2} \sum_{r \geq 0} \frac{g_r(j+1/2)}{n^r}.$$

To obtain the coefficient of $n^{-k-3/2}$, set $r = k - j$ and sum over $0 \leq j \leq k$. Finally use

$$\Gamma\left(-j - \frac{1}{2}\right) = \frac{(-4)^{j+1}(j+1)!\sqrt{\pi}}{(2j+2)!}$$

to obtain Equation (6.11). Standard singularity analysis in a Δ-domain at the unique dominant singularity gives the stated remainder at every fixed order [11, 13]. □

The first normalized transfer identities are particularly simple:

$$\Theta_0 = -\frac{1}{2} t_1, \tag{6.12}$$

$$\Theta_1 = -\frac{3}{16} t_1 + \frac{3}{4} t_3, \tag{6.13}$$

$$\Theta_2 = -\frac{25}{256} t_1 + \frac{45}{32} t_3 - \frac{15}{8} t_5, \tag{6.14}$$

$$\Theta_3 = -\frac{105}{2048} t_1 + \frac{1155}{512} t_3 - \frac{525}{64} t_5 + \frac{105}{16} t_7, \tag{6.15}$$

$$\Theta_4 = -\frac{1659}{65536} t_1 + \frac{14175}{4096} t_3 - \frac{26775}{1024} t_5 + \frac{6615}{128} t_7 - \frac{945}{32} t_9. \tag{6.16}$$

6.3 The leading Otter constant

Since $t_1 = -\sqrt{2\ell_1}$,

$$A_0 = -\frac{t_1}{2\sqrt{\pi}} = \sqrt{\frac{\ell_1}{2\pi}} = \sqrt{\frac{1+\rho R'(\rho)}{2\pi}}. \tag{6.17}$$

Numerically,

$$A_0 = 0.439924012571025304040903391434544764798085407940 12\ldots. \tag{6.18}$$

Thus the classical first-order form is

$$a_n \sim 0.4399240125710253040409\ldots\ (2.9557652856519949747148\ldots)^n n^{-3/2}. \tag{6.19}$$

6.4 Numerical coefficient table

It is convenient to display $\Theta_k = \sqrt{\pi}A_k$, so that

$$a_n \sim \frac{\rho^{-n}}{\sqrt{\pi n^3}} \sum_{k\geq 0} \frac{\Theta_k}{n^k}. \tag{6.20}$$

Table 6.1: Forward coefficients in Equation (6.20).

k	Θ_k
0	0.7797450101873204427711036979
1	0.07828911261061096206127535867
2	0.3929402676631860184743155979
3	1.537879315978838091102498124
4	8.200844090435596159219524856
5	57.29291473494343787739929540
6	503.0445050262735818247740217
7	5359.600933884326033403142719
8	67342.06920114653048340562550
9	975425.4970695924732213025531
10	15996932.49293173312960557140
11	292822531.3353392231145687742
12	5914523441.293932519117404924

Factoring out A_0 gives

$$a_n \sim A_0\alpha^n n^{-3/2} F(n^{-1}), \qquad F(u) = 1 + \sum_{k\geq 1} c_k u^k, \qquad c_k = \frac{A_k}{A_0} = \frac{\Theta_k}{\Theta_0}. \tag{6.21}$$

The first relative coefficients are

$$\begin{aligned} c_1 &= 0.1004034800964014347637283944\ldots, \\ c_2 &= 0.5039343151023035331021858170\ldots, \\ c_3 &= 1.972284908382278469205675365\ldots, \\ c_4 &= 10.51734090413157399273764936\ldots, \\ c_5 &= 73.47647498402047106416577934\ldots, \\ c_6 &= 645.1397552456612928153464923\ldots. \end{aligned} \tag{6.22}$$

6.5 Numerical validation and optimal truncation

Table 6.2 compares exact integers from Equation (1.6) with truncations after n^{-K}. The entries are signed relative errors $(a_n^{[K]} - a_n)/a_n$.

Table 6.2: Signed relative errors of the forward expansion.

n	$K=0$	$K=1$	$K=2$	$K=4$	$K=6$	$K=8$	$K=10$
20	$-6.58\,10^{-3}$	$-1.59\,10^{-3}$	$-3.42\,10^{-4}$	$-3.18\,10^{-5}$	$1.04\,10^{-6}$	$9.73\,10^{-6}$	$1.41\,10^{-5}$
50	$-2.22\,10^{-3}$	$-2.19\,10^{-4}$	$-1.77\,10^{-5}$	$-2.88\,10^{-7}$	$-1.20\,10^{-8}$	$-9.84\,10^{-10}$	$-1.36\,10^{-10}$
100	$-1.06\,10^{-3}$	$-5.24\,10^{-5}$	$-2.08\,10^{-6}$	$-8.06\,10^{-9}$	$-7.88\,10^{-11}$	$-1.50\,10^{-12}$	$-4.73\,10^{-14}$
200	$-5.15\,10^{-4}$	$-1.28\,10^{-5}$	$-2.53\,10^{-7}$	$-2.40\,10^{-10}$	$-5.73\,10^{-13}$	$-2.66\,10^{-15}$	$-2.04\,10^{-17}$
500	$-2.03\,10^{-4}$	$-2.03\,10^{-6}$	$-1.59\,10^{-8}$	$-2.39\,10^{-12}$	$-9.02\,10^{-16}$	$-6.62\,10^{-19}$	$-8.01\,10^{-22}$

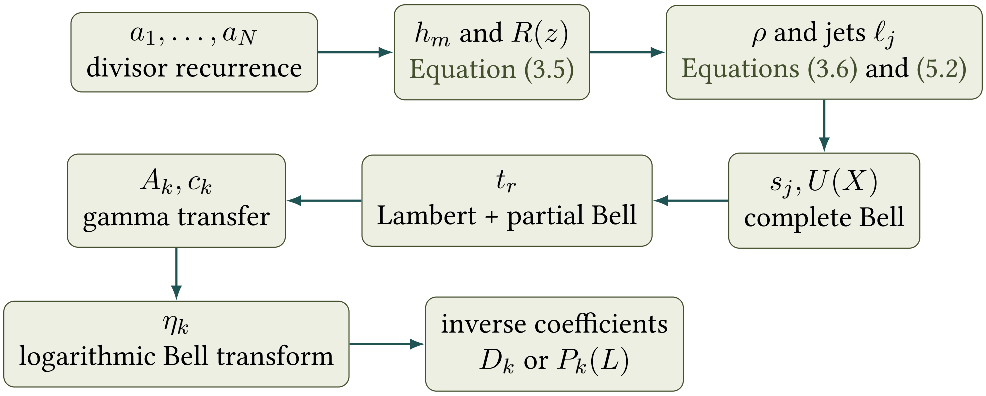
At fixed n the expansion is asymptotic, not convergent: the improvement stops when the growing coefficients overcome the powers of n. For example, at $n = 20$ the sixth-order truncation is better than the eighth- or tenth-order truncation. This late-term growth is the perturbative shadow of farther singularities, which are made explicit in Part IV.

Chapter 7

A reproducible coefficient pipeline

7.1 Dependency graph

The coefficient calculation is triangular in the following sense.



No high-order nonlinear system is solved. At every stage, the coefficient of order r depends only on data of order at most r.

7.2 Exact integer stage

The following Python-like pseudocode realizes Equation (1.6). The array b is updated as a divisor sieve.

Listing 7.1: Exact generation of A000081.

```
def rooted_tree_numbers(N):
    a = [0] * (N + 1)
    b = [0] * (N + 1)    # b[m] = sum_{d | m} d*a[d]

    a[1] = 1
    for m in range(1, N + 1):
        b[m] += 1        # contribution of d = 1

    for n in range(2, N + 1):
        numerator = sum(b[j] * a[n-j] for j in range(1, n))
        assert numerator % (n - 1) == 0
        a[n] = numerator // (n - 1)

        for m in range(n, N + 1, n):
            b[m] += n * a[n]
```

```
    return a, b
```

The coefficients of R are then exactly

$$h_m = \frac{b_m - m a_m}{m}. \tag{7.1}$$

In fact h_m is rational, though the defining proper-divisor sum often makes its denominator cancellation transparent.

7.3 Finding ρ and its jets

For a target precision of P decimal digits, choose N so that $(\sqrt{\rho})^N$ is safely below 10^{-P}, compute $a_1, \ldots, a_N$, and solve

$$f(x) = \log x + 1 + \sum_{m=2}^{N} h_m x^m = 0 \tag{7.2}$$

by Newton or safeguarded secant iteration. The omitted tail is geometrically small because R is analytic in $|z| < \sqrt{\rho}$. Once ρ is known, Equation (5.3) evaluates all required jets by rapidly convergent sums.

7.4 Formal-series stage

For requested forward order K, it is enough to compute

1. $\ell_1, \ldots, \ell_{K+1}$;
2. $s_1, \ldots, s_{K+1}$ by Equation (5.5);
3. $\omega_1, \ldots, \omega_{2K+1}$ by Equation (4.3);
4. $t_1, t_3, \ldots, t_{2K+1}$ by Equation (5.8);
5. $g_r(j + 1/2)$ for $j + r \leq K$ by Equation (6.5);
6. $A_0, \ldots, A_K$ by Equation (6.10).

Naive polynomial arithmetic costs $O(K^3)$ scalar operations. Bell recurrences, FFT multiplication, and relaxed power-series methods reduce this when hundreds or thousands of coefficients are required. For the orders normally useful in numerical asymptotics, the naive triangular implementation is simpler and more reliable.

Precision discipline

The asymptotic coefficients eventually grow rapidly. A calculation of A_K with d reliable digits may require substantially more than d working digits because Equation (6.10) combines alternating contributions from several Puiseux coefficients. The calculation underlying the tables used 90–120 decimal working digits and exact rational arithmetic for the universal coefficients and gamma-ratio coefficients.

Part III

The Asymptotic Inverse

Chapter 8

What it means to invert a sequence

8.1 Three inverse objects

A sequence is not literally a real-variable function, and A000081 begins with $a_1 = a_2 = 1$. The phrase "inverse of the sequence" can therefore denote three different objects.

Definition 8.1 (Smooth asymptotic inverse)**.** Let

$$\mathcal{A}(x) = A_0 \alpha^x x^{-3/2} F(x^{-1}) \tag{8.1}$$

be the formal asymptotic interpolation obtained from Equation (6.21). Its asymptotic inverse $\nu(y)$ is the formal solution of $\mathcal{A}(\nu(y)) = y$ with $\nu(y) \to +\infty$.

Definition 8.2 (Generalized integer inverse)**.** For $y > 1$, define

$$N_*(y) = \min\{n \geq 2 : a_n \geq y\}. \tag{8.2}$$

This is a right-continuous staircase.

Definition 8.3 (Exact interpolated inverse)**.** Choose a monotone interpolation $\mathcal{A}_{\text{ex}}(x)$ satisfying $\mathcal{A}_{\text{ex}}(n) = a_n$; for example, interpolate $\log a_n$ linearly on each interval $[n, n+1]$. Let ν_{ex} be its ordinary inverse.

The smooth inverse is canonical at the level of asymptotic series. The exact interpolated inverse depends on the chosen interpolation between integers, and the generalized integer inverse retains an order-one rounding oscillation. These distinctions do not affect the coefficients derived below, but they do affect claims of pointwise accuracy smaller than one.

8.2 The unavoidable rounding layer

For nonintegral x,

$$\lceil x \rceil = x + \frac{1}{2} + \frac{1}{\pi} \sum_{m \geq 1} \frac{\sin(2\pi m x)}{m}, \tag{8.3}$$

with the usual midpoint interpretation at integers. Thus, once an exact monotone interpolation has been chosen,

$$N_*(y) = \lceil \nu_{\text{ex}}(y) \rceil. \tag{8.4}$$

The Fourier series in Equation (8.3) is not a small correction: it is an $O(1)$ periodic layer. Consequently no smooth function can approximate $N_*(y)$ uniformly with error $o(1)$ for all real y. At the special values $y = a_n$, however, the smooth inverse satisfies $\nu(a_n) = n + o(1)$, and successive asymptotic orders make this error arbitrarily small before the divergent late-term regime is reached.

Important qualification

An expansion such as $N_*(y) = \nu(y) + o(1)$ is false uniformly in y. The correct statement is either about the smooth inverse ν, about values $y = a_n$, or about a rounded exact interpolation. The integer staircase and the analytic inverse must not be silently identified.

Chapter 9

Normalization of the inverse problem

9.1 Logarithmic phase

Set

$$p = \frac{3}{2}, \qquad C = A_0, \qquad \lambda = \log \alpha = -\log \rho. \tag{9.1}$$

Starting from $y = C\mathrm{e}^{\lambda x} x^{-p} F(x^{-1})$, introduce

$$X = \lambda x, \qquad Z = \log \frac{y}{C\lambda^p}. \tag{9.2}$$

Then

$$Z = X - p \log X + K(X^{-1}), \tag{9.3}$$

where

$$K(v) = \log F(\lambda v) = \sum_{k \geq 1} \eta_k v^k. \tag{9.4}$$

The logarithm is the key simplification: the multiplicative forward correction becomes an additive inverse-power phase.

9.2 General coefficient formula for the phase

By Equation (2.7),

$$\boxed{\eta_n = \frac{1}{n!} \sum_{k=1}^{n} (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1! c_1 \lambda, 2! c_2 \lambda^2, \ldots).} \tag{9.5}$$

In particular,

$$\begin{aligned}
\eta_1 &= \lambda c_1, \\
\eta_2 &= \lambda^2 \left(c_2 - \frac{1}{2} c_1^2 \right), \\
\eta_3 &= \lambda^3 \left(c_3 - c_1 c_2 + \frac{1}{3} c_1^3 \right), \\
\eta_4 &= \lambda^4 \left(c_4 - c_1 c_3 - \frac{1}{2} c_2^2 + c_1^2 c_2 - \frac{1}{4} c_1^4 \right).
\end{aligned} \tag{9.6}$$

Numerically,

$$\begin{aligned}
\eta_1 &= 0.1088130343442745145167554784\ldots, \\
\eta_2 &= 0.5859660997717814776889432009\ldots, \\
\eta_3 &= 2.4465585734567796498814798 60\ldots, \\
\eta_4 &= 14.0675323511249824230302 3266\ldots, \\
\eta_5 &= 106.854661153452369127111 9951\ldots, \\
\eta_6 &= 1022.1764672232478727675 88518\ldots.
\end{aligned} \tag{9.7}$$

Also

$$C\lambda^p = 0.4963361416595997474727369464252 2249134814\ldots. \tag{9.8}$$

Chapter 10

Lambert–W core and an ordinary correction series

10.1 Exact summation of the logarithmic skeleton

First omit K and solve

$$X_0 - p \log X_0 = Z. \tag{10.1}$$

The large positive solution is

$$\boxed{X_0 = -pW_{-1}\left(-\frac{1}{p}\mathrm{e}^{-Z/p}\right).} \tag{10.2}$$

Indeed, with $r = X_0/p$, $r\mathrm{e}^{-r} = \mathrm{e}^{-Z/p}/p$, and the large solution corresponds to the branch W_{-1}. This single expression resums the complete tower $Z + p\log Z + p^2 \log Z/Z + \cdots$ generated by the power-law prefactor x^{-p}.

10.2 Correction about the exact core

Write

$$X = X_0 + \delta. \tag{10.3}$$

Subtracting Equation (10.1) from Equation (9.3) gives

$$\delta = p \log\left(1 + \frac{\delta}{X_0}\right) - K\left(\frac{1}{X_0 + \delta}\right). \tag{10.4}$$

Set $\varepsilon = X_0^{-1}$ and define

$$\Phi_\varepsilon(u) = p\log(1 + \varepsilon u) - K\left(\frac{\varepsilon}{1 + \varepsilon u}\right). \tag{10.5}$$

Then $\delta = \Phi_\varepsilon(\delta)$ and $\Phi_\varepsilon = O(\varepsilon)$.

Theorem 10.1 (All inverse corrections about the Lambert core). *As a formal series in X_0^{-1},*

$$\boxed{\delta = \sum_{m\geq 1} \frac{1}{m!}\, \frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}} \Phi_\varepsilon(u)^m \Big|_{u=0}, \qquad \varepsilon = X_0^{-1}.} \tag{10.6}$$

Equivalently,

$$\delta \sim \sum_{r\geq 1} \frac{D_r}{X_0^r}, \tag{10.7}$$

where the completely explicit coefficient extractor is

$$\boxed{D_r = [\varepsilon^r] \sum_{m=1}^{r} \frac{1}{m!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}} \left[p \log(1+\varepsilon u) - K\left(\frac{\varepsilon}{1+\varepsilon u}\right)\right]^m \Bigg|_{u=0}.} \tag{10.8}$$

Finally,

$$\boxed{\nu(y) \sim \frac{1}{\lambda}\left(X_0 + \sum_{r\geq 1} \frac{D_r}{X_0^r}\right), \quad X_0 = -pW_{-1}\left(-\frac{1}{p}\left(\frac{C\lambda^p}{y}\right)^{1/p}\right).} \tag{10.9}$$

Proof. Introduce an auxiliary marker t and solve $\delta = t\Phi_\varepsilon(\delta)$. Lagrange–Bürmann gives

$$\delta(t) = \sum_{m\geq 1} \frac{t^m}{m!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}} \Phi_\varepsilon(u)^m \Bigg|_{u=0}.$$

Since $\Phi_\varepsilon = O(\varepsilon)$, the coefficient of ε^r receives contributions only from $m \leq r$, so setting $t = 1$ is formally legitimate and gives Equation (10.8). Division by λ returns from X to x. ☐

10.3 First correction coefficients

The first D_r are

$$\begin{aligned}
D_1 &= -\eta_1, && (10.10)\\
D_2 &= -p\eta_1 - \eta_2, && (10.11)\\
D_3 &= -p^2\eta_1 - p\eta_2 - \eta_1^2 - \eta_3, && (10.12)\\
D_4 &= -p^3\eta_1 - p^2\eta_2 - \frac{5}{2}p\eta_1^2 - p\eta_3 - 3\eta_1\eta_2 - \eta_4, && (10.13)\\
D_5 &= -p^4\eta_1 - p^3\eta_2 - \frac{9}{2}p^2\eta_1^2 - p^2\eta_3 - 7p\eta_1\eta_2 - p\eta_4 \\
&\quad - 2\eta_1^3 - 4\eta_1\eta_3 - 2\eta_2^2 - \eta_5. && (10.14)
\end{aligned}$$

For A000081 these are

$$\begin{aligned}
D_1 &= -0.10881303434427451451675547 84\ldots,\\
D_2 &= -0.74918565128819324946407641 85\ldots,\\
D_3 &= -3.5821773268322777891030098 44\ldots,\\
D_4 &= -19.658721211387760037075563 94\ldots,\\
D_5 &= -138.53274785185608613136349 30\ldots,\\
D_6 &= -1248.9793263716928544965065 50\ldots,\\
D_7 &= -13905.408844748148354198909 12\ldots,\\
D_8 &= -184719.19112650809048944593 91\ldots.
\end{aligned} \tag{10.15}$$

The coefficients grow, as expected for an asymptotic series; W_{-1} has already removed the dominant logarithmic combinatorics, but it cannot remove the farther singularities encoded in the forward coefficients.

Chapter 11

Polynomial–logarithmic inverse transseries

11.1 Direct expansion without a special function

Some applications prefer an expansion involving only elementary logarithms. Let

$$L = \log Z, \qquad X = Z + U, \qquad U = \sum_{n\geq 0} P_n(L) Z^{-n}. \tag{11.1}$$

With $\varepsilon = Z^{-1}$, substitution into Equation (9.3) gives a single generating equation for all coefficient polynomials:

$$\boxed{P(\varepsilon, L) = pL + p\log\big(1 + \varepsilon P(\varepsilon, L)\big) - K\left(\frac{\varepsilon}{1 + \varepsilon P(\varepsilon, L)}\right),} \tag{11.2}$$

where $P(\varepsilon, L) = \sum_{n\geq 0} P_n(L)\varepsilon^n$.

11.2 A closed triangular recurrence

Define convolution polynomials

$$C_{m,r}(L) = [\varepsilon^m] P(\varepsilon, L)^r = \sum_{j_1+\cdots+j_r=m} P_{j_1}(L)\cdots P_{j_r}(L), \tag{11.3}$$

with $C_{0,0} = 1$ and $C_{m,0} = 0$ for $m > 0$. Expanding the logarithm and the negative powers in Equation (11.2) yields the following all-order recurrence.

Theorem 11.1 (Coefficient polynomials of the direct inverse). *One has $P_0(L) = pL$, and for $n \geq 1$,*

$$\boxed{\begin{aligned} P_n(L) = p\sum_{r=1}^{n} \frac{(-1)^{r+1}}{r} C_{n-r,r}(L) \\ - \sum_{k=1}^{n} \eta_k \sum_{r=0}^{n-k} (-1)^r \binom{k+r-1}{r} C_{n-k-r,r}(L) \end{aligned}} \tag{11.4}$$

Moreover P_n is a polynomial of degree n for $n \geq 1$, and

$$\boxed{\nu(y) \sim \frac{1}{\lambda}\left[Z + p\log Z + \sum_{n\geq 1} \frac{P_n(\log Z)}{Z^n}\right], \quad Z = \log\frac{y}{C\lambda^p}.} \tag{11.5}$$

Proof. Use

$$\log(1+\varepsilon P) = \sum_{r\geq 1} \frac{(-1)^{r+1}}{r} \varepsilon^r P^r$$

and

$$K\left(\frac{\varepsilon}{1+\varepsilon P}\right) = \sum_{k\geq 1} \eta_k \varepsilon^k \sum_{r\geq 0} (-1)^r \binom{k+r-1}{r} \varepsilon^r P^r.$$

Extracting ε^n gives Equation (11.4). The right side at order n uses only $P_0, \ldots, P_{n-1}$, so the recurrence is triangular. Induction on n proves the degree bound, and substitution establishes the asymptotic expansion. □

11.3 First coefficient polynomials

The first five are

$$P_0(L) = pL, \tag{11.6}$$
$$P_1(L) = p^2 L - \eta_1, \tag{11.7}$$
$$P_2(L) = -\frac{1}{2} p^3 L^2 + (p^3 + p\eta_1) L - p\eta_1 - \eta_2, \tag{11.8}$$
$$\begin{aligned} P_3(L) = {} & \frac{1}{3} p^4 L^3 - \left(\frac{3}{2} p^4 + p^2 \eta_1\right) L^2 \\ & + \left(p^4 + 3p^2 \eta_1 + 2p\eta_2\right) L - p^2 \eta_1 - p\eta_2 - \eta_1^2 - \eta_3, \end{aligned} \tag{11.9}$$
$$\begin{aligned} P_4(L) = {} & -\frac{1}{4} p^5 L^4 + \left(\frac{11}{6} p^5 + p^3 \eta_1\right) L^3 \\ & - \left(3p^5 + \frac{11}{2} p^3 \eta_1 + 3p^2 \eta_2\right) L^2 \\ & + \left(p^5 + 6p^3 \eta_1 + 5p^2 \eta_2 + 3p\eta_1^2 + 3p\eta_3\right) L \\ & - p^3 \eta_1 - p^2 \eta_2 - \frac{5}{2} p\eta_1^2 - p\eta_3 - 3\eta_1 \eta_2 - \eta_4. \end{aligned} \tag{11.10}$$

Notice that

$$D_n = P_n(0). \tag{11.11}$$

This is not accidental: the Lambert core sums exactly the part of the direct recurrence generated by pL; the remaining constants obey the same fixed-point calculus with $L = 0$.

For $p = 3/2$ and the A000081 phase coefficients,

$$\begin{aligned} P_0(L) = {} & 1.5L, \\ P_1(L) = {} & 2.25L - 0.10881303434427451452\ldots, \\ P_2(L) = {} & -1.6875L^2 + 3.5382195515164117718\ldots L \\ & - 0.74918565128819324946\ldots, \\ P_3(L) = {} & 1.6875L^3 - 7.8385793272746176577\ldots L^2 \\ & + 7.5548862811391974061\ldots L - 3.5821773268322777891\ldots, \\ P_4(L) = {} & -1.8984375L^4 + 14.289118990911926486\ldots L^3 \\ & - 28.756363123475120650\ldots L^2 \\ & + 27.452127392454046160\ldots L - 19.658721211387760037\ldots. \end{aligned} \tag{11.12}$$

11.4 Elementary leading form

Expanding only to constant accuracy in terms of $Y = \log y$ gives

$$\nu(y) = \frac{\log y}{\lambda} + \frac{p}{\lambda} \log\log y - \frac{1}{\lambda} \log(C\lambda^p) + o(1). \tag{11.13}$$

Numerically,

$$\begin{aligned}\nu(y) = {} & 0.92271556161860152480\ldots \log y \\ & + 1.38407334242790228720\ldots \log\log y \\ & + 0.64636398270020880729\ldots + o(1).\end{aligned} \tag{11.14}$$

The constant in this coarser form depends on the exact normalization of Z; using Equation (11.5) is preferable whenever more than the first two scales are needed.

Chapter 12

Accuracy, implementation, and integer recovery

12.1 Comparison at exact sequence values

For $y = a_n$, define the order-K Lambert-core approximation

$$\nu^{[K]}(a_n) = \frac{1}{\lambda}\left(X_0 + \sum_{r=1}^{K} \frac{D_r}{X_0^r}\right). \tag{12.1}$$

Table 12.1 gives the signed index error $\nu^{[K]}(a_n) - n$.

Table 12.1: Signed error of the asymptotic inverse at $y = a_n$.

n	$K=0$	$K=1$	$K=2$	$K=4$	$K=6$	$K=8$	$K=10$
20	$6.55\,10^{-3}$	$1.91\,10^{-3}$	$4.44\,10^{-4}$	$3.76\,10^{-5}$	$-2.02\,10^{-7}$	$-9.39\,10^{-6}$	$-1.39\,10^{-5}$
50	$2.11\,10^{-3}$	$2.59\,10^{-4}$	$2.32\,10^{-5}$	$3.32\,10^{-7}$	$1.26\,10^{-8}$	$9.98\,10^{-10}$	$1.35\,10^{-10}$
100	$9.88\,10^{-4}$	$6.16\,10^{-5}$	$2.74\,10^{-6}$	$9.34\,10^{-9}$	$8.35\,10^{-11}$	$1.53\,10^{-12}$	$4.71\,10^{-14}$
200	$4.78\,10^{-4}$	$1.50\,10^{-5}$	$3.33\,10^{-7}$	$2.79\,10^{-10}$	$6.09\,10^{-13}$	$2.71\,10^{-15}$	$2.03\,10^{-17}$
500	$1.88\,10^{-4}$	$2.38\,10^{-6}$	$2.10\,10^{-8}$	$2.78\,10^{-12}$	$9.59\,10^{-16}$	$6.75\,10^{-19}$	$7.98\,10^{-22}$

The inverse error is essentially the forward relative error divided by the logarithmic slope $\lambda - p/n + O(n^{-2})$. This explains the close numerical parallel between Tables 6.2 and 12.1.

12.2 A robust numerical procedure

For a large positive target y, a stable implementation is:

Algorithm 12.1 (Asymptotic inversion of A000081)**.**

1. Compute $Z = \log(y/(C\lambda^{3/2}))$ using a logarithmic representation of y if y is too large for ordinary floating point.
2. Evaluate
$$X_0 = -\frac{3}{2}W_{-1}\left(-\frac{2}{3}\mathrm{e}^{-2Z/3}\right).$$
3. Choose a truncation order K and form $X = X_0 + \sum_{r=1}^{K} D_r X_0^{-r}$.
4. Return $x = X/\lambda$. If an integer index is desired, test the two or three integers nearest to x by exact recurrence values, arbitrary precision logarithms, or a certified forward bound.

A single Newton correction can be applied to a chosen smooth truncation:

$$x_{\text{new}} = x - \frac{\log \mathcal{A}_K(x) - \log y}{\lambda - p/x + \frac{\mathrm{d}}{\mathrm{d}x} \log F_K(x^{-1})}. \tag{12.2}$$

Working with logarithms avoids overflow and loss of relative precision.

12.3 Rounding certification

Because $a_{n+1}/a_n \to \alpha > 1$, adjacent sequence values eventually differ by a fixed multiplicative factor. Hence an index estimate with absolute error less than $1/2$ identifies the nearest integer at targets $y = a_n$. For a general target y, certification of $N_*(y)$ requires checking

$$a_{m-1} < y \leq a_m. \tag{12.3}$$

The asymptotic inverse reduces this check to a constant-size neighbourhood of $m \approx \nu(y)$; it does not replace the endpoint convention in the definition of the staircase.

Part IV

Beyond All Orders: The Singularity Transseries

Chapter 13

The recursive singularity atlas

13.1 Why one exponential scale is not the whole story

The expansion in Equation (6.9) is complete in inverse powers of n at the dominant exponential scale ρ^{-n}. It does not contain terms such as σ^{-n} with $|\sigma| > \rho$, because those are smaller than every power of $1/n$ relative to ρ^{-n}. Such terms arise from farther singularities of analytic continuation. For the Pólya-tree equation there are infinitely many of them, and they accumulate at the unit circle, which is a natural boundary [10].

The functional equation exposes two mechanisms.

1. **Critical Lambert singularities.** On a branch where R is analytic, a point σ satisfying $\zeta(\sigma) = 1/e$ is a square-root singularity whenever $\zeta'(\sigma) \neq 0$.
2. **Inherited singularities.** If T is singular at τ and $\sigma^k = \tau$ for some $k \geq 2$, then the term $T(z^k)/k$ can make $R(z)$, hence $T(z)$, singular at σ. Repeated root lifts produce singularities whose moduli approach one and whose arguments become dense.

Different paths can reach different determinations of the inner functions and of the outer Lambert function. The appropriate global object is therefore the Riemann surface of the germ T at zero, not a single-valued function on the whole punctured unit disk.

13.2 Local closure of the Puiseux calculus

Fix a continuation sheet and a point $\sigma \neq 0$. Choose a local uniformizer v such that

$$z = \sigma(1 - v^M) \tag{13.1}$$

for some positive integer M, and suppose all inner germs needed in $R(z)$ have Puiseux expansions in v. Algebraic addition, composition with z^k, exponentiation, and multiplication preserve the ring $\mathbb{C}[[v]]$. Thus

$$\zeta(v) = \zeta_0 + \sum_{m \geq 1} \zeta_m v^m. \tag{13.2}$$

There are two cases.

13.2.1 Regular outer composition

Let C_b denote the branch of the Cayley function reached on the chosen sheet, and assume

$$y_0 = C_b(\zeta_0) \neq 1. \tag{13.3}$$

Then C_b is analytic at ζ_0, and

$$T(v) = y_0 + \sum_{m\geq 1} y_m v^m, \qquad \boxed{y_m = \frac{1}{m!}\sum_{k=1}^{m} C_b^{(k)}(\zeta_0)\mathcal{B}_{m,k}(1!\zeta_1, 2!\zeta_2, \ldots).} \tag{13.4}$$

The derivatives are explicit functions of y_0. Since $u = y\mathrm{e}^{-y}$,

$$C_b'(u) = \frac{\mathrm{e}^y}{1-y}, \qquad C_b^{(k)}(u) = \left[\left(\frac{\mathrm{e}^y}{1-y}\frac{\mathrm{d}}{\mathrm{d}y}\right)^{k-1}\frac{\mathrm{e}^y}{1-y}\right]_{y=C_b(u)}. \tag{13.5}$$

13.2.2 Critical outer composition

If $\zeta_0 = 1/e$ and the reached branch has $y_0 = 1$, define

$$Q(v) = \sqrt{2(1 - e\zeta(v))}. \tag{13.6}$$

Then the universal formula of Theorem 4.1 gives

$$\boxed{T(v) = 1 - \sum_{m\geq 1}\omega_m Q(v)^m.} \tag{13.7}$$

If $1 - e\zeta(v)$ vanishes first at order r, then Q begins with $v^{r/2}$; the ramification index doubles when r is odd. This is the only new ramification mechanism. Consequently the complete singularity atlas can be generated recursively by ordinary formal power-series arithmetic and the same universal sequence (ω_m).

Theorem 13.1 (Recursive completeness of the local calculus)**.** *Starting from the dominant germ at ρ, every isolated singular germ reached by finitely many applications of the root-lift and critical mechanisms admits a computable Puiseux expansion. At each fixed local order, its coefficients are finite expressions built from*

1. *previously computed local coefficients of inner germs;*
2. *complete and partial Bell polynomials;*
3. *the rational Lambert coefficients ω_m;*
4. *derivatives of a regular Cayley branch as in Equation (13.5).*

Proof. Proceed by induction on the depth of the singularity in the root-lift graph. At depth zero, Equation (5.8) gives the dominant germ. At the induction step, substitution $z \mapsto z^k$ converts every already known Puiseux germ to a Puiseux series in a common local uniformizer. The sum defining R, its exponential, and multiplication by z preserve that ring. If the outer Cayley branch is regular, apply Equation (13.4); if it is critical, apply Equation (13.7). Both formulae determine every coefficient at finite order. □

13.3 Contour-resolved coefficient transseries

Let a chosen deformation of the Cauchy coefficient contour encounter isolated singularities σ on specified sheets. Suppose the local singular part is

$$T_\sigma(z) \sim_{\mathrm{asy}} \sum_{q\in E_\sigma} c_{\sigma,q}\left(1 - \frac{z}{\sigma}\right)^q, \tag{13.8}$$

where $E_\sigma \subset \mathbb{Q}_{\geq 0}$ and analytic exponents $q \in \mathbb{N}_0$ may be omitted from the local Hankel contribution. Let $\mathfrak{S}_\sigma$ be the contour intersection or Stokes multiplier associated with that local branch.

Full contour-resolved forward transseries

Under the usual finite-radius continuation and remainder hypotheses for the chosen contour,

$$a_n \sim_{\text{asy}} \sum_{\sigma} \mathfrak{S}_\sigma\, \sigma^{-n} \sum_{q \in E_\sigma \setminus \mathbb{N}_0} \frac{c_{\sigma,q}}{\Gamma(-q)} n^{-q-1} \sum_{r \geq 0} \frac{g_r(q)}{n^r}. \tag{13.9}$$

Relative to the dominant sector, define the action

$$\mathfrak{A}_\sigma = \operatorname{Log} \frac{\sigma}{\rho}. \tag{13.10}$$

Then $\sigma^{-n} = \rho^{-n} \mathrm{e}^{-n\mathfrak{A}_\sigma}$, so Equation (13.9) is a genuine exponential transseries. Conjugate singularities combine into real oscillatory sectors.

Coefficient-contour argument. Deform the Cauchy contour from a small circle to a finite union of local Hankel contours about the encountered singularities plus an outer remainder contour. Apply Equation (6.1) termwise to each truncated local Puiseux expansion and then use Theorem 6.1. The outer contour gives the radius-dependent exponential remainder. Passing through a branch cut or to a different sheet changes the homology coefficients of the local Hankel cycles; these are precisely the multipliers $\mathfrak{S}_\sigma$. □

Remark 13.2 (Projective meaning of the infinite sum). Because singularities accumulate at $|z| = 1$, the sum over σ is not meant as an absolutely convergent unordered numerical series. For every fixed continuation radius $R < 1$, only the singularities met before the R-contour are included, and the rest is absorbed into the outer remainder. The global transseries is the compatible family of these finite-radius expansions as the contour is enlarged. This is the natural notion of fullness in the presence of a natural boundary.

Chapter 14

The first inherited sector at $-\sqrt{\rho}$

14.1 A singularity accessible along the negative axis

Let

$$\sigma_- = -\sqrt{\rho} = -0.58165441363339425381005778164483483128 33\ldots. \tag{14.1}$$

The germ at the origin can be continued along the negative real axis without meeting the dominant positive singularity. At $z = \sigma_-$, the term $T(z^2)/2$ in $R(z)$ reaches $T(\rho)/2$ and inherits the dominant square-root branch. All terms $T(z^k)$ with $k \geq 3$ are evaluated strictly inside the original disk of convergence because $|\sigma_-|^k < \rho$.

Choose

$$z = \sigma_-(1 - v^2), \qquad v = \left(1 - \frac{z}{\sigma_-}\right)^{1/2}. \tag{14.2}$$

Then

$$z^2 = \rho(1 - v^2)^2, \qquad 1 - \frac{z^2}{\rho} = v^2(2 - v^2).$$

Using the dominant coefficients t_r, the complete local disturbance is

$$\begin{aligned} R_-(v) = \frac{1}{2}&\left[1 + \sum_{r\geq 1} t_r v^r (2 - v^2)^{r/2}\right] \\ &+ \sum_{k\geq 3} \frac{1}{k} \sum_{n\geq 1} a_n \sigma_-^{kn} (1 - v^2)^{kn}. \end{aligned} \tag{14.3}$$

Both lines are convergent as local Puiseux/power series near $v = 0$.

14.2 General coefficients of the inherited disturbance

Write $R_-(v) = \sum_{m\geq 0} r_m^- v^m$ and put $t_0 = 1$. For $j = (m - r)/2$, direct coefficient extraction from Equation (14.3) gives

$$\begin{aligned} r_m^- = \frac{1}{2} &\sum_{\substack{0\leq r\leq m \\ r\equiv m \pmod 2}} t_r (-1)^j 2^{r/2-j} \binom{r/2}{j} \\ &+ \mathbf{1}_{2|m} (-1)^{m/2} \sum_{k\geq 3} \frac{1}{k} \sum_{n\geq 1} a_n \sigma_-^{kn} \binom{kn}{m/2}. \end{aligned} \tag{14.4}$$

This is an explicit all-order formula. The double sum in the second line converges geometrically, already at $k = 3$.

Define

$$\zeta_-(v) = \sigma_-(1-v^2)\mathrm{e}^{R_-(v)} = \sum_{m\geq 0} z_m^- v^m. \tag{14.5}$$

If

$$E_m^- = \frac{\mathrm{e}^{r_0^-}}{m!}\mathcal{B}_m(1!r_1^-, 2!r_2^-, \ldots, m!r_m^-), \qquad E_m^- = 0 \quad (m<0), \tag{14.6}$$

then

$$z_m^- = \sigma_-(E_m^- - E_{m-2}^-). \tag{14.7}$$

Finally let

$$y_- = C_0(z_0^-) = -W_0(-z_0^-), \qquad T(\sigma_-(1-v^2)) = y_- + \sum_{m\geq 1} b_m^- v^m. \tag{14.8}$$

The outer Cayley branch is regular because $y_- \neq 1$, so

$$\boxed{b_m^- = \frac{1}{m!}\sum_{k=1}^{m} C_0^{(k)}(z_0^-)\mathcal{B}_{m,k}(1!z_1^-, 2!z_2^-, \ldots).} \tag{14.9}$$

Together, Equations (14.4), (14.7) and (14.9) are the complete coefficient calculus for this first inherited sector.

14.3 Leading coefficient in closed form

The coefficient of v in R_- comes only from $T(z^2)/2$:

$$r_1^- = \frac{t_1}{\sqrt{2}}. \tag{14.10}$$

Since $z_1^- = z_0^- r_1^-$ and $C_0'(z_0^-) = y_-/[z_0^-(1-y_-)]$,

$$\boxed{b_1^- = \frac{y_-}{1-y_-}\frac{t_1}{\sqrt{2}}.} \tag{14.11}$$

Numerically,

$$\begin{aligned}
r_0^- &= 0.46863662792484554808720260335458271132 34\ldots, \\
z_0^- &= -0.92937573520120508885013672457275632261 86\ldots, \\
y_- &= -0.54103294142045349262050309866364171642 82\ldots, \\
b_1^- &= 0.38715011103024290347720669374748831003 49\ldots.
\end{aligned} \tag{14.12}$$

The first local coefficients are

$$\begin{aligned}
b_1^- &= 0.387150111030242903477206 6937\ldots, \\
b_2^- &= -0.032564918515647634147309 81111\ldots, \\
b_3^- &= 0.017200421603122091483007 54229\ldots, \\
b_4^- &= 0.104077039518384438525489 6204\ldots, \\
b_5^- &= 0.161947360772702361189504 1812\ldots, \\
b_6^- &= -0.062925998187993086897986 87888\ldots, \\
b_7^- &= -0.244031039231421351354611 5758\ldots.
\end{aligned} \tag{14.13}$$

14.4 The alternating coefficient sector

Only odd b^-_{2j+1} are singular in z at σ_-. The same gamma-ratio transfer as before gives

$$a_n^{(-)} \sim_{\text{asy}} \mathfrak{S}_- \sigma_-^{-n} n^{-3/2} \sum_{k \geq 0} \frac{B_k^-}{n^k}, \tag{14.14}$$

where

$$\boxed{B_k^- = \sum_{j=0}^{k} \frac{b^-_{2j+1}}{\Gamma(-j-1/2)} g_{k-j}\left(j + \frac{1}{2}\right).} \tag{14.15}$$

For the direct negative-axis continuation the local Hankel multiplier is one; we have retained $\mathfrak{S}_-$ in Equation (14.14) to make the contour dependence explicit in a larger, multi-sheet deformation.

Since $\sigma_-^{-n} = (-1)^n \rho^{-n/2}$, the numerical local sector is

$$\begin{aligned} a_n^{(-)} \sim_{\text{asy}} &\ \mathfrak{S}_-(-1)^n \rho^{-n/2} n^{-3/2} \\ &\times \Bigg(-0.1092130299563101757054885230\ldots \\ &\qquad - \frac{0.03367666220778285336719302163\ldots}{n} \\ &\qquad - \frac{0.1790009011730463163890152410\ldots}{n^2} \\ &\qquad - \frac{1.642342004953279983002352205\ldots}{n^3} \\ &\qquad - \frac{18.92664586329208396709681008\ldots}{n^4} - \cdots \Bigg). \end{aligned} \tag{14.16}$$

Relative to the leading dominant term, its first amplitude is

$$\frac{B_0^-}{A_0} = -0.2482543049151649529915710045\ldots, \qquad \frac{a_n^{(-)}}{a_n^{(0)}} \sim -0.2482543049\ldots\,(-\sqrt{\rho})^n. \tag{14.17}$$

This is smaller than every power of $1/n$, and hence invisible to the complete series Equation (6.9); it is the first explicit genuinely transseries correction.

Remark 14.1 (Other sectors of the same or greater modulus). The point $+\sqrt{\rho}$ lies behind the dominant branch point on real continuation and belongs to sheet-dependent continuations. Complex critical points and overlapping root lifts may also occur farther out. Their local coefficients are generated by Theorem 13.1; their global multipliers depend on the selected contour. Formula Equation (14.16) is therefore an explicit sector, not an assertion that one term alone exhausts every continuation at modulus $\sqrt{\rho}$.

Chapter 15

Transport of exponential sectors through inversion

15.1 General transseries setup

Let the full forward interpolation be written as

$$\mathcal{A}_{\text{full}}(x) = \mathcal{A}_0(x)\big(1 + \mathcal{E}(x)\big), \tag{15.1}$$

where $\mathcal{A}_0$ contains the complete dominant inverse-power sector and $\mathcal{E}$ is a well-ordered sum of exponentially small sectors,

$$\mathcal{E}(x) = \sum_{\omega\in\Omega\setminus\{0\}} \mathrm{e}^{-\omega x} x^{\mu_\omega} \sum_{r\geq 0} e_{\omega,r} x^{-r}. \tag{15.2}$$

Here Ω is generated by the singular actions $\mathfrak{A}_\sigma = \mathrm{Log}(\sigma/\rho)$. Taking a logarithm gives

$$\mathcal{R}(x) = \log(1 + \mathcal{E}(x)) = \sum_{m\geq 1} \frac{(-1)^{m+1}}{m} \mathcal{E}(x)^m. \tag{15.3}$$

Products add actions, so the inverse transseries is supported on the additive semigroup generated by Ω. Coefficients at any fixed action and inverse power are finite Cauchy products, equivalently multivariate Bell-polynomial expressions.

Let

$$f_0(x) = \log \mathcal{A}_0(x), \qquad f_0(x_0) = \log y, \tag{15.4}$$

where x_0 is the complete perturbative inverse of Parts III, either in the W_{-1} form or the polynomial–logarithmic form. Put $x = x_0 + \Delta$ and

$$A_{x_0}(u) = \begin{cases} \dfrac{f_0(x_0+u) - f_0(x_0)}{u}, & u \neq 0, \\ f_0'(x_0), & u = 0. \end{cases} \tag{15.5}$$

Then the exact formal inversion equation is

$$\Delta = \Psi_{x_0}(\Delta), \qquad \Psi_{x_0}(u) = -\frac{\mathcal{R}(x_0+u)}{A_{x_0}(u)}. \tag{15.6}$$

Theorem 15.1 (All-sector inverse Lagrange formula). *The beyond-all-orders correction to the inverse is*

$$\boxed{\Delta = \sum_{m\geq 1} \frac{1}{m!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}u^{m-1}} \Psi_{x_0}(u)^m \bigg|_{u=0}.} \tag{15.7}$$

When Equation (15.3) and $A_{x_0}^{-1}$ are expanded formally, this formula generates every mixed action, every power of x_0^{-1}, and every polynomial factor arising from differentiation.

Proof. Equation Equation (15.6) is of the form required by the formal Lagrange–Bürmann theorem. Introduce an auxiliary marker t, solve $\Delta = t\Psi_{x_0}(\Delta)$, apply Equation (2.6), and set $t = 1$. Well ordering of the action support ensures that each requested transmonomial receives only finitely many contributions. $\square$

15.2 First-order transport law

The first term of Equation (15.7) is

$$\Delta = -\frac{\mathcal{R}(x_0)}{f_0'(x_0)} + O(\mathcal{R}^2). \tag{15.8}$$

Since $f_0'(x_0) = \lambda - p/x_0 + O(x_0^{-2})$, a relative forward sector

$$\mathcal{E}_\omega(x) \sim e_{\omega,0}\mathrm{e}^{-\omega x}x^{\mu_\omega} \tag{15.9}$$

produces

$$\Delta_\omega(y) \sim -\frac{e_{\omega,0}}{\lambda}\mathrm{e}^{-\omega x_0}x_0^{\mu_\omega}. \tag{15.10}$$

Using $y \sim C\mathrm{e}^{\lambda x_0}x_0^{-p}$,

$$\boxed{\Delta_\omega(y) \sim -\frac{e_{\omega,0}}{\lambda}C^{\omega/\lambda}y^{-\omega/\lambda}x_0^{\mu_\omega - p\omega/\lambda}.} \tag{15.11}$$

Complex actions therefore become complex powers of y and oscillations in $\log y$, with an additional $\log\log y$ phase modulation through x_0.

15.3 Inverse image of the $-\sqrt{\rho}$ sector

For the inherited negative sector,

$$\omega_- = \operatorname{Log}\frac{\sigma_-}{\rho} = \frac{\lambda}{2} + \mathrm{i}\pi \tag{15.12}$$

(up to the lateral choice of $\pm\pi$), and $e_{-,0} = B_0^-/A_0 = -0.2482543049\ldots$. A real median interpolation of the integer alternation replaces $\mathrm{e}^{-\mathrm{i}\pi x}$ by $\cos(\pi x)$, giving

$$\begin{aligned}\Delta_-(y) &\sim -\frac{B_0^-}{\lambda A_0}\mathrm{e}^{-\lambda x_0/2}\cos(\pi x_0)\\ &= 0.229068110384031978565646715 4\ldots\ \mathrm{e}^{-\lambda x_0/2}\cos(\pi x_0).\end{aligned} \tag{15.13}$$

Since $\mathrm{e}^{-\lambda x_0/2} \sim C^{1/2}y^{-1/2}x_0^{-3/4}$,

$$\boxed{\Delta_-(y) \sim 0.151933473659530988262729039 1\ldots\ y^{-1/2}x_0^{-3/4}\cos(\pi x_0).} \tag{15.14}$$

The leading phase is

$$\pi x_0 = \frac{\pi}{\lambda}\log y + \frac{3\pi}{2\lambda}\log\log y + O(1), \qquad \frac{\pi}{\lambda} = 2.898796429733978737710727372\ldots. \tag{15.15}$$

Thus the first nonperturbative inverse correction has magnitude $y^{-1/2}(\log y)^{-3/4}$ and a logarithmic oscillation whose phase contains a slower $\log\log y$ drift.

15.4 Higher corrections and mixed actions

The next orders follow mechanically from Equation (15.7):

1. inverse powers in the ω_- sector come from $B_1^-, B_2^-, \ldots$, from the complete perturbative derivative $f_0'(x_0)$, and from differentiation of $\mathrm{e}^{-\omega_- u}$;
2. the two-instanton sector $2\omega_-$ is produced by the quadratic term in $\log(1+\mathcal{E})$ and by the $m = 2$ Lagrange term;
3. products with any other singularity action produce sectors $r\omega_- + \omega_{\sigma_1} + \cdots + \omega_{\sigma_s}$;
4. conjugate actions combine into real trigonometric polynomials in the corresponding logarithmic phases.

At each fixed action and inverse order the calculation is finite. This is the precise sense in which Equation (15.7) supplies general formulae for the coefficients of the full inverse transseries.

Two distinct oscillatory phenomena

The exponentially small oscillation in Equation (15.14) comes from a complex singular action and tends to zero. The order-one sawtooth in Equation (8.3) comes from integer rounding and never tends to zero. They have different origins, different scales, and must not be conflated.

Chapter 16

Summary of the main formulae

For reference, the complete calculation may be compressed to the following chain.

1. Generate a_n and b_n exactly from
$$(n-1)a_n = \sum_{j=1}^{n-1} b_j a_{n-j}, \qquad b_j = \sum_{d|j} d a_d.$$
2. Set
$$R(z) = \sum_{m\geq 2} \frac{b_m - m a_m}{m} z^m, \qquad \log\rho + 1 + R(\rho) = 0.$$
3. Form the scaled jets
$$\ell_j = (-1)^{j-1}(j-1)! + \sum_{m\geq j} \frac{b_m - m a_m}{m} m^{\underline{j}} \rho^m.$$
4. Compute
$$s_j = \frac{2(-1)^{j+1}}{j!}\mathcal{B}_j(\ell_1,\ldots,\ell_j), \qquad U(X) = \frac{\sum_{j\geq 1} s_j X^j}{s_1 X}.$$
5. Use the universal coefficients
$$\omega_m = \frac{1}{m}[u^{m-1}]\left(\sum_{j\geq 0} \frac{2u^j}{(j+2)j!}\right)^{-m/2}$$
and
$$t_r = -\sum_{\substack{m\leq r\\ m\equiv r\ (2)}} \omega_m s_1^{m/2}[X^{(r-m)/2}]U(X)^{m/2}.$$
6. Transfer to sequence coefficients:
$$A_k = \sum_{j=0}^{k} \frac{t_{2j+1}}{\Gamma(-j-1/2)} g_{k-j}(j+1/2),$$
where g_r is generated from Bernoulli-polynomial coefficients Equations (6.2) and (6.5).
7. Normalize $c_k = A_k/A_0$ and compute
$$\eta_n = [v^n]\log\left(1 + \sum_{k\geq 1} c_k \lambda^k v^k\right)$$
by Equation (9.5).

8. Invert with either

$$\nu(y) \sim_{\text{asy}} \frac{1}{\lambda}\left(X_0 + \sum_{r\geq 1}\frac{D_r}{X_0^r}\right), \quad X_0 = -\frac{3}{2}W_{-1}\left(-\frac{2}{3}\left(\frac{C\lambda^{3/2}}{y}\right)^{2/3}\right),$$

where D_r is given by Equation (10.8), or with the polynomial–logarithmic recurrence Equation (11.4).

9. Generate all farther singular sectors recursively by Equations (13.4) and (13.7), transfer each by Equation (13.9), and invert all sectors by Equation (15.7).

Final dominant pair

The forward and inverse expansions, in their most useful normalized forms, are

$$a_n \sim_{\text{asy}} C\alpha^n n^{-3/2}\left(1 + \sum_{k\geq 1}\frac{c_k}{n^k}\right),$$
$$\nu(y) \sim_{\text{asy}} \frac{1}{\lambda}\left[Z + \frac{3}{2}\log Z + \sum_{n\geq 1}\frac{P_n(\log Z)}{Z^n}\right], \qquad Z = \log\frac{y}{C\lambda^{3/2}},$$

with C, α, c_k, and P_n determined by finite coefficient formulae above. The first nonperturbative sector is proportional to $(-1)^n\rho^{-n/2}n^{-3/2}$ forward and to $y^{-1/2}(\log y)^{-3/4}$ times a logarithmic oscillation after inversion.

Appendix A

Fully explicit finite coefficient formulae

The main text organizes the calculation by mathematical dependency. This appendix removes the remaining shorthand and displays the coefficient operators as finite sums. Its purpose is not to replace the more readable formulae in the body, but to make the phrase "general formula for the coefficients" completely literal.

A.1 Bell-polynomial recurrences

The partial exponential Bell polynomials satisfy

$$\mathcal{B}_{n,k}(x_1, x_2, \ldots) = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j \mathcal{B}_{n-j,k-1}(x_1, x_2, \ldots), \tag{A.1}$$

with $\mathcal{B}_{0,0} = 1$ and zero outside $0 \leq k \leq n$. Thus all Bell quantities in this article can be generated by a triangular array without enumerating partitions. The complete polynomials obey

$$\mathcal{B}_n(x_1, \ldots, x_n) = \sum_{j=1}^{n} \binom{n-1}{j-1} x_j \mathcal{B}_{n-j}(x_1, \ldots, x_{n-j}), \qquad \mathcal{B}_0 = 1. \tag{A.2}$$

Both recurrences follow by distinguishing the block containing a fixed marked atom in the set-partition interpretation.

For a unit series $U(X) = 1 + \sum_{j \geq 1} u_j X^j$, introduce

$$\mathfrak{P}_N(\beta; u_1, \ldots, u_N) := \frac{1}{N!} \sum_{k=0}^{N} \beta^{\underline{k}} \, \mathcal{B}_{N,k}(1!u_1, 2!u_2, \ldots, (N-k+1)!u_{N-k+1}). \tag{A.3}$$

Then $[X^N]\, U(X)^\beta = \mathfrak{P}_N(\beta; u_1, \ldots, u_N)$. In particular, every fractional power used below is an explicitly finite polynomial in the input coefficients and in β.

A.2 Universal Lambert coefficients as a partition sum

Recall

$$H(u) = \sum_{j \geq 0} \frac{2u^j}{(j+2)j!} = 1 + \sum_{j \geq 1} h_j^{\mathrm{L}} u^j, \qquad h_j^{\mathrm{L}} = \frac{2}{(j+2)j!}. \tag{A.4}$$

The superscript L distinguishes these universal Lambert coefficients from the Pólya-disturbance coefficients h_j in Equation (3.5). Combining Lagrange inversion with Equation (A.3) gives

$$\boxed{\omega_m = \frac{1}{m(m-1)!} \sum_{k=0}^{m-1} -m/2^{\underline{k}} \, \mathcal{B}_{m-1,k}\big(1!h_1^{\mathrm{L}}, 2!h_2^{\mathrm{L}}, \ldots\big).} \tag{A.5}$$

This proves at once that $\omega_m \in \mathbb{Q}$ for every m.

For convenient checking, the first eighteen values are

Table A.1: Exact universal coefficients in $1 - T = \sum_{m\geq 1} \omega_m q^m$.

m	ω_m	m	ω_m
1	1	10	$-5776369/1515591000$
2	$-1/3$	11	$169709463197/69528040243200$
3	$11/72$	12	$-1118511313/709296588000$
4	$-43/540$	13	$667874164916771/650782456676352000$
5	$769/17280$	14	$-500525573/744761417400$
6	$-221/8505$	15	$103663334225097487/234281684403486720000$
7	$680863/43545600$	16	$-466901817532379/1595278956070800000$
8	$-1963/204120$	17	$21235294185086305043/109242202556140093440000$
9	$226287557/37623398400$	18	$-106040742894306601/818378104464320400000$

A.3 One composite formula for every dominant forward coefficient

Let

$$u_j = \frac{s_{j+1}}{s_1}, \qquad \mathfrak{P}_N(\beta) = \mathfrak{P}_N(\beta; u_1, \ldots, u_N). \tag{A.6}$$

Then Equations (5.8) and (6.10) combine to give

$$A_k = -\sum_{j=0}^{k} \frac{g_{k-j}(j+1/2)}{\Gamma(-j-1/2)} \sum_{r=0}^{j} \omega_{2r+1} s_1^{r+1/2} \mathfrak{P}_{j-r}(r+1/2). \tag{A.7}$$

No infinite formal sum occurs on the right: for fixed k it contains finitely many rational operations after the analytic jets $s_1, \ldots, s_{k+1}$ have been supplied. Those jets themselves are explicit convergent sums:

$$\ell_j = (-1)^{j-1}(j-1)! + \sum_{m\geq j} \frac{b_m - m a_m}{m} m^{\underline{j}} \rho^m, \tag{A.8}$$

$$s_j = \frac{2(-1)^{j+1}}{j!} \mathcal{B}_j(\ell_1, \ldots, \ell_j). \tag{A.9}$$

Consequently Equations (A.7) to (A.9), together with the exact divisor recurrence Equation (1.6), are a complete formula for A_k involving no fitted constants.

Remark A.1 (Rational and analytic layers). The numbers ω_m and the polynomials $g_r(\beta)$ are universal and rational. All dependence on the particular unlabelled-tree class enters through ρ and the disturbance jets ℓ_j. This separation is useful both conceptually and computationally: the universal arrays may be precomputed once and reused for any tree class satisfying $T = \zeta e^T$ with analytic ζ.

A.4 Completely expanded logarithmic phase coefficients

Writing $f_j = c_j \lambda^j$, the additive phase coefficients are

$$\boxed{\eta_n = \frac{1}{n!} \sum_{k=1}^{n} (-1)^{k-1}(k-1)! \mathcal{B}_{n,k}(1! f_1, 2! f_2, \ldots).} \tag{A.10}$$

Equivalently, by expanding the Bell polynomial into multiplicities,

$$\eta_n = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_j m_j = k,\ \sum_j j m_j = n}} \frac{k!}{\prod_j m_j!} \prod_{j\geq 1} (c_j \lambda^j)^{m_j}. \tag{A.11}$$

This is the ordinary logarithm formula grouped by integer partitions of n.

A.5 A finite partition formula for the Lambert-core corrections

Let

$$\Phi_\varepsilon(u) = p\log(1+\varepsilon u) - \sum_{k\geq 1} \eta_k \left(\frac{\varepsilon}{1+\varepsilon u}\right)^k, \qquad p = \frac{3}{2}. \tag{A.12}$$

The coefficient D_r in Equation (10.8) may be written without derivatives as

$$\boxed{D_r = \sum_{m=1}^{r} \frac{1}{m}[\varepsilon^r u^{m-1}]\,\Phi_\varepsilon(u)^m.} \tag{A.13}$$

For each r, truncate Φ_ε at total ε-degree r; then the coefficient in Equation (A.13) is a finite multinomial sum. This is often the simplest exact implementation because bivariate truncated series multiplication automatically performs all Faà di Bruno bookkeeping.

Appendix B

Gamma-ratio transfer to all orders

B.1 The exact singular-monomial coefficient

For $\beta \notin \mathbb{N}_0$ and $n \geq 0$,

$$[z^n]\left(1-\frac{z}{\rho}\right)^{\beta} = \rho^{-n}(-1)^n\binom{\beta}{n} = \rho^{-n}\frac{\Gamma(n-\beta)}{\Gamma(-\beta)\Gamma(n+1)}. \tag{B.1}$$

If $\beta \in \mathbb{N}_0$, the left side vanishes for $n > \beta$. This explains why the integer powers $t_{2j}X^j$ in the local expansion do not contribute to the algebraic coefficient asymptotics: they are locally analytic and have no Hankel jump.

B.2 Bernoulli-polynomial derivation

For fixed $a, b \in \mathbb{C}$, Euler–Maclaurin expansion of $\log\Gamma$ gives

$$\log\frac{\Gamma(n+a)}{\Gamma(n+b)n^{a-b}} \sim_{\text{asy}} \sum_{m\geq 1}\frac{(-1)^{m+1}}{m(m+1)}\frac{B_{m+1}(a)-B_{m+1}(b)}{n^m}, \tag{B.2}$$

where $B_m(x)$ denotes the Bernoulli polynomial. Setting $a=-\beta$ and $b=1$ yields the coefficients $d_m(\beta)$ of Equation (6.2). Exponentiating then gives both

$$g_r(\beta) = \frac{1}{r!}\mathcal{B}_r(1!d_1(\beta), 2!d_2(\beta), \ldots, r!d_r(\beta)) \tag{B.3}$$

and the triangular recurrence Equation (6.5).

The first five polynomials are

$$\begin{aligned}
g_0(\beta) &= 1,\\
g_1(\beta) &= \frac{\beta(\beta+1)}{2},\\
g_2(\beta) &= \frac{\beta(\beta+1)(\beta+2)(3\beta+1)}{24},\\
g_3(\beta) &= \frac{\beta^2(\beta+1)^2(\beta+2)(\beta+3)}{48},\\
g_4(\beta) &= \frac{\beta(\beta+1)(\beta+2)(\beta+3)(\beta+4)(15\beta^3+30\beta^2+5\beta-2)}{5760}.
\end{aligned} \tag{B.4}$$

In particular $g_r(\beta)$ is a polynomial of degree $2r$ with rational coefficients.

B.3 Remainder form of the transfer theorem

Let $M \geq 0$ be fixed. Suppose that, in a Δ-domain at ρ,

$$T(z) = H(z) + \sum_{j=0}^{M} t_{2j+1} \left(1 - \frac{z}{\rho}\right)^{j+1/2} + \mathcal{O}\left(\left|1 - \frac{z}{\rho}\right|^{M+3/2}\right), \tag{B.5}$$

where H is analytic in a disk of radius strictly larger than ρ after the local analytic terms are patched away. Then

$$a_n = \rho^{-n} n^{-3/2} \left(\sum_{k=0}^{M} \frac{A_k}{n^k} + \mathcal{O}(n^{-M-1})\right) + \mathcal{O}(R^{-n}) \tag{B.6}$$

for some $R > \rho$ determined by the chosen continuation domain. The first error is the algebraic truncation error at the dominant singularity; the second records all farther singularities not crossed by the contour. This separation is precisely what is refined into the transseries of Equation (13.9).

Proof. Apply the exact identity Equation (B.1) to each displayed singular monomial and truncate its gamma-ratio expansion Equation (B.2). A Hankel-contour estimate gives $\rho^{-n} n^{-M-5/2}$ for the local remainder. Cauchy's estimate on the remaining outer contour gives $O(R^{-n})$. This is the standard transfer mechanism of singularity analysis [11, 12]. □

Appendix C

Structure and error theory of the inverse expansion

C.1 Existence and uniqueness in the logarithmic scale

Consider the general phase

$$Z = X - p \log X + K(X^{-1}), \qquad K(v) = \sum_{j \geq 1} \eta_j v^j, \qquad p > 0. \tag{C.1}$$

The derivative is

$$\frac{\mathrm{d}Z}{\mathrm{d}X} = 1 - \frac{p}{X} - \frac{1}{X^2} K'(X^{-1}) = 1 + O(X^{-1}). \tag{C.2}$$

Hence the map is strictly increasing for all sufficiently large positive X, and it has a unique inverse germ at $+ealsymbol$.

Theorem C.1 (Formal uniqueness). *There is a unique logarithmic transseries of the form*

$$X = Z + p \log Z + \sum_{n \geq 1} P_n(\log Z) Z^{-n} \tag{C.3}$$

that solves Equation (C.1) coefficientwise. Each P_n is a polynomial with coefficients in $\mathbb{Q}[p, \eta_1, \ldots, \eta_n]$.

Proof. Substitute Equation (C.3), expand $\log X = \log Z + \log(1 + (X - Z)/Z)$ and expand $K(X^{-1})$ as a unit-series composition. At order Z^{-n} the unknown P_n occurs linearly with coefficient one, while all other terms involve only $P_0, \ldots, P_{n-1}$. Thus the coefficients are determined triangularly. The operations used are addition, multiplication, formal logarithm, and unit-series powers, so they preserve the stated coefficient ring. □

C.2 Degree, leading term, and constant term

The triangular recurrence contains additional structure.

Proposition C.2 (Shape of the inverse polynomials). *For $n \geq 1$,*

$$\deg P_n = n, \qquad [L^n] P_n(L) = \frac{(-1)^{n+1} p^{n+1}}{n}, \qquad P_n(0) = D_n. \tag{C.4}$$

Here D_n is the Lambert-core coefficient from Equation (10.8).

Proof. The highest power of L is independent of K: every occurrence of an η_j consumes at least one inverse power without increasing the maximum logarithmic degree. Thus the leading term is the same as for $Z = X - p\log X$. Expanding its exact inverse $X = -pW_{-1}(-e^{-Z/p}/p)$, or inducting directly in Equation (11.4), gives $(-1)^{n+1}p^{n+1}L^n/n$.

For the constant term, set $L = 0$ in the direct recurrence. The resulting near-identity reversion is exactly the correction equation about the Lambert core with X_0^{-1} as small parameter. Formal uniqueness therefore identifies its order-n coefficient with D_n. ☐

C.3 Transport of a controlled forward remainder

Suppose a positive smooth interpolant has logarithm

$$f(x) = \lambda x - p\log x + \log C + \sum_{j=1}^{M} \kappa_j x^{-j} + O(x^{-M-1}), \qquad \lambda > 0. \tag{C.5}$$

Let $x_M(y)$ be the inverse obtained by matching the displayed expansion through order M. Since $f'(x) = \lambda + O(x^{-1})$, the mean-value theorem gives

$$x_M(y) - f^{-1}(\log y) = O\big((\log y)^{-M-1}\big)\,. \tag{C.6}$$

More precisely, the leading residual is divided by $f'(x_M) = \lambda - p/x_M + O(x_M^{-2})$. This explains why forward relative errors and inverse absolute errors in Table 12.1 have nearly the same asymptotic order, apart from the factor $1/\lambda$.

C.4 Staircase recovery and the best possible deterministic error

Let ν be any continuous asymptotic inverse agreeing with a monotone interpolation through the sequence values. Then

$$N_*(y) = \lceil \nu(y) + \epsilon(y) \rceil, \qquad \epsilon(y) \to 0, \tag{C.7}$$

provided y stays away from the finitely many initial ties. No continuous asymptotic series can determine the integer answer with an error tending to zero uniformly in y: arbitrarily small changes of y across a_n change $N_*(y)$ by one. The Fourier sawtooth in Equation (8.3) is therefore not an optional refinement but the canonical order-one transseries layer of the generalized inverse.

Appendix D

Numerical audit and reusable constants

D.1 Dominant singular constants

The constants used in the tables are

$$\begin{aligned}
\rho &= 0.338321856899207695196112625717017053183774607532967795572303776 3\ldots, \\
\alpha &= 2.9557652856519949747148175241231945883754923046635965953504724 79\ldots, \\
\lambda &= 1.0837575972446246604827119808961742790496621242026722349144270 65\ldots, \\
C = A_0 &= 0.43992401257102530404090339143454476479808540794012\ldots, \\
C\lambda^{3/2} &= 0.4963361416595997474727369464252249134 8\ldots.
\end{aligned} \tag{D.1}$$

They satisfy the independent checks

$$\lambda = -\log \rho = 1 + R(\rho), \qquad C^2 = \frac{1 + \rho R'(\rho)}{2\pi}. \tag{D.2}$$

D.2 Local Puiseux coefficients

For $X = 1 - z/\rho$, $T(z) = 1 + \sum_{r \geq 1} t_r X^{r/2}$ begins as follows.

Table D.1: Numerical dominant Puiseux coefficients.

r	t_r	r	t_r
1	$-1.5594900203746408855422073 96$	10	$0.2059848312779074187361657348$
2	$0.8106697078826992814381417462$	11	$-0.0527247084981905623153382 1627$
3	$-0.2854870216128456053038513708$	12	$0.0170265687549536694460453950 2$
4	$0.1653723657120838934765135390$	13	$-0.1523706243663253962860946421$
5	$-0.3424599704021542010422002117$	14	$0.1737028832998504628939446698$
6	$0.3174072259465285635680708874$	15	$-0.0144737037395270448567311125 0$
7	$-0.1077788002916310091822310895$	16	$-0.0218995176112155622326728374 7$
8	$0.0613849570558351046887314992 5$	17	$-0.1445471935709097054769461137$
9	$-0.1952123835975564636127698720$	18	$0.1760771088850177790623322866$

The alternating, nonmonotone pattern is expected: these are local analytic coefficients, not the final coefficient-asymptotic constants, and even-indexed terms do not directly survive singularity transfer.

D.3 Inverse phase and Lambert-core coefficients

The first additive phase coefficients η_j and correction coefficients D_j are

Table D.2: Coefficients in $K(v) = \sum \eta_j v^j$ and $\delta(X_0) = \sum D_j X_0^{-j}$.

j	η_j	D_j
1	0.1088130343442745145167554784	−0.1088130343442745145167554784
2	0.5859660997717814776889432009	−0.7491856512881932494640764185
3	2.446558573456779649881479860	−3.582177326832123...
4	14.06753235112498242303023266	−19.658721211387...
5	106.8546611534523691271119951	−138.532747851856...
6	1022.176467223247872767588518	−1248.97932637169...
7	11859.388275...	−13905.4088447481...
8	162087.382638...	−184719.191126508...

The growth visible here is one reason to retain the exact W_{-1} core: it sums the entire logarithmic skeleton before the rapidly growing correction series is truncated.

D.4 Exact integer benchmarks

The divisor recurrence gives, without floating-point arithmetic,

n	a_n
1	1
10	719
20	12 826 228
30	354 426 847 597
40	11 703 780 079 612 453
50	425 976 989 835 141 038 353
75	135 410 903 503 191 503 503 384 705 970 501
100	51 384 328 351 659 326 880 337 136 395 054 298 255 277 970

These values provide direct regression tests for any implementation of Equation (1.6) and for the error tables in the main text.

Appendix E

Notation and scale map

Several variables deliberately carry similar letters because each normalizes a different stage. The following table collects them in one place.

Table E.1: Principal symbols and their roles.

Symbol	Meaning
a_n	Number of unlabelled non-plane rooted trees on n vertices; A000081.
$T(z)$	Ordinary generating function $\sum_{n\geq 1} a_n z^n$.
b_m	Divisor transform $\sum_{d\mid m} d a_d$.
$R(z)$	Analytic disturbance $\sum_{k\geq 2} T(z^k)/k$.
$\zeta(z)$	Cayley argument $z \exp R(z)$, so $T = -W_0(-\zeta)$.
ρ	Positive dominant singularity; $T(\rho) = 1$.
α, λ	$\alpha = \rho^{-1}$ and $\lambda = \log \alpha$.
$X = 1 - z/\rho$	Local forward singular coordinate.
q	Universal Lambert coordinate $\sqrt{2(1 - e\zeta(z))}$.
ω_m	Rational inverse coefficients in $1 - T = \sum \omega_m q^m$.
t_r	Puiseux coefficients in $T = 1 + \sum t_r X^{r/2}$.
A_k, c_k	Absolute and relative dominant coefficient-asymptotic constants.
x	Continuous large index in the smooth interpolation $\mathcal{A}(x)$.
$X = \lambda x$	Dimensionless inverse index; context distinguishes it from the local forward X.
Z	Normalized logarithm $\log(y/(C\lambda^p))$.
L	Slow logarithm $\log Z$.
$K(v)$	Additive correction phase $\log F(\lambda v)$.
X_0	Exact Lambert core $-pW_{-1}(-e^{-Z/p}/p)$.
D_r	Ordinary inverse-power corrections about X_0.
$P_r(L)$	Polynomial–logarithmic coefficients in the direct inverse expansion.
$\sigma, \mathfrak{A}_\sigma$	A farther singularity and its action $\mathrm{Log}(\sigma/\rho)$.
$\mathfrak{S}_\sigma$	Contour intersection/Stokes multiplier for a chosen continuation sheet.

Bibliographic note

Pólya's cycle-index method and Otter's analysis supply the classical rooted-tree foundation. The Cayley-disturbance viewpoint and full dominant expansion are closest in spirit to Genitrini's treatment, while the explicit coefficient organization here follows the Bell-polynomial and reversion calculus developed in the linked ProveIt drafts. The all-order inverse, the distinction between smooth and staircase inverses, the recursively contour-resolved farther singularity sectors, and the explicit transport of those sectors through the inverse are developed here as a unified extension of those ingredients.

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